RESEARCH & PROTOTYPE REVIEW

Prototype Feature Inventory

What appears in the supplied app export and which claims still require verification.

SIH 2026 | PS SIH26250 | Alchemy Architects

SOURCE-CODE REVIEW + PUBLIC RESEARCH | 20 OCTOBER 2026

ALCHEMY ARCHITECTS

Feature inventory with evidence labels

This inventory is based on the supplied AI Studio source ZIP and screenshots, not a successful live browser inspection. Categories mean: SOURCE PRESENT = component or logic exists; SEEDED/STATIC = shown with prefilled data; LIVE-CONNECTOR-CODE = a request path exists but runtime success is not certified; NOT FOUND = implementation not found in the reviewed source paths.

Application areas

Area	What the export appears to provide	Evidence label / caveat
Command Centre	Dashboard tiles, data connector status, readiness summary, alerts and 3D resource constellation.	SOURCE PRESENT; KPI source integrity must be checked.
Readiness Registry	Resource states, capability tags, status and source labels.	SOURCE PRESENT; examples include seeded records.
Workforce / equipment	Dedicated views for workforce readiness and equipment/inventory.	SOURCE PRESENT; dataset quality and sensitivity need review.
Data Quality Studio	Stale/unknown/conflict issue panel and scanner concept.	SOURCE PRESENT; validate each detector with labelled errors.
Planning Sandbox	Priority-based deterministic heuristic, configurable weights, scheduled assignments and baseline comparison.	ALGORITHM PRESENT; not ML/CP-SAT; deadline/objective gaps need tests.
Independent validator	Readiness, availability, maintenance, overlap, capability, dependency and freshness warnings.	RULES PRESENT; does not visibly cover all necessary constraints.
Dynamic Replanning	Controlled outage/weather/maintenance-delay scenario presets and re-solve.	SOURCE PRESENT; uses demo records, not real-world response validation.
Weather/Public Data	AviationWeather.gov connector code and IMD-style display/fallback path.	CONNECTOR CODE PRESENT; exact live data state not verified.
Predictive Maintenance	NASA C-MAPSS label, five sample records and RUL table.	STATIC/SEEDED; no reproducible trained model found.
Model Evaluation	Fixed baseline/candidate RMSE/MAE/interval and sensor weights.	STATIC; displayed values not substantiated by training script.
VAYU NEXA AI	Server-side Gemini request path and rule-based offline fallback.	CODE PRESENT; live model IDs, secrets, grounding and response quality unverified.
Evidence Ledger	Session audit list and downloadable evidence concepts.	SOURCE PRESENT; integrity/persistence and server security require validation.
Requirements Coverage	35-item traceability matrix and status labels.	SOURCE PRESENT; self-declared labels must be backed by tests.
3D/visual views	Three.js constellation, scenario comparison and asset/visual modules.	SOURCE PRESENT; public safe demo should keep only generic, non-operational visuals.
Firebase	Client configuration and Firestore rule file.	CONFIG PRESENT; original broad rules require replacement before use.

Features that should not be claimed as delivered without more proof

- A trained ML/DL model or verified +56.7% RUL error reduction.
- A proven CP-SAT/global optimum solver. The source uses a custom deterministic heuristic.
- Fully live or independently verified IMD/NASA/BTS feeds simply because a source card says “LIVE”.
- Verified military fleet/crew readiness, operational airbase status, or authorised access to defence datasets.
- Grounded real-time AI answers unless successful API requests, model identifiers and returned source citations are captured and tested.

- A visit or endorsement by an airport, IMD, AAI, DSSC, MoD or any other office without dated evidence and permission.

References and source notes

[1] AI Studio application URL
https://aakaash-adhinayak.ai.studio/
User-supplied hosted prototype link. Automated retrieval was unsuccessful in this review; screenshots and the exported source bundle were inspected instead.

[2] AviationWeather.gov Data API
https://aviationweather.gov/data/api/
Official public civil aviation weather data API documentation; connector status and terms must be verified before live claims.

[3] NASA Prognostics Center of Excellence Dataset Repository
https://www.nasa.gov/intelligent-systems-division/discovery-and-systems-health/pcoe/pcoe-data-set-repository/
Official NASA repository lists turbofan engine degradation simulation data and the C-MAPSS benchmark. It is not military telemetry.